AI Value Stack

Chips → cloud → models → apps

PROFESSIONAL DEFINITION

The AI Value Stack is a layered map of the AI economy — semiconductors, cloud infrastructure, foundation models, and applications — showing where cost is incurred and where value and margin accrue.

WHO DEVELOPED IT

A synthesis from venture and industry analysis of the generative-AI supply chain, discussed by investors such as Andreessen Horowitz and Sequoia.

WHY IT WAS DEVELOPED

It was developed to help firms see that AI cost and value are distributed across a stack, and to locate where they are exposed or where they capture margin.

FIGURE 1 · AI VALUE STACK

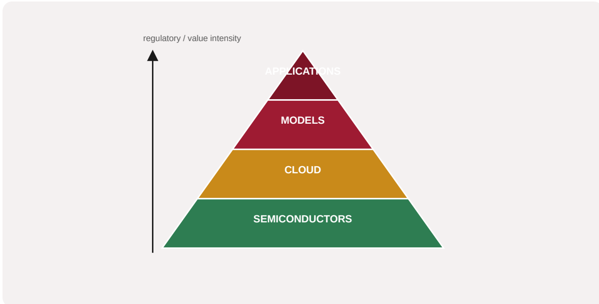

Figure 1. A schematic of the AI Value Stack as applied in BARBM312 (illustrative).

HOW IT IS USED

A firm maps its dependencies and spend across the layers; concentration in any layer signals cost or lock-in risk to be managed.

WHEN IT IS USED

When designing AI sourcing strategy and analysing cost structure and dependency.

BY WHOM IT IS USED

CTOs, strategists and finance leaders.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

It gives a single picture of the AI economy's layers, clarifying that most application-layer firms rent the layers beneath them.

RELATED FRAMEWORKS IN THIS COURSE

- **Compute Cost Pyramid** · Slibrary 9
- **Token Economics** · Slibrary 9
- **Hyperscaler Dependency Map** · Slibrary 9
- **Capex vs. Opex Shift** · Slibrary 9

SOURCES (HARVARD REFERENCING STYLE)

a16z (2023) 'Who owns the generative AI platform?'.
Sequoia Capital (2023) 'Generative AI: A creative new world'.